

Numerical Methods

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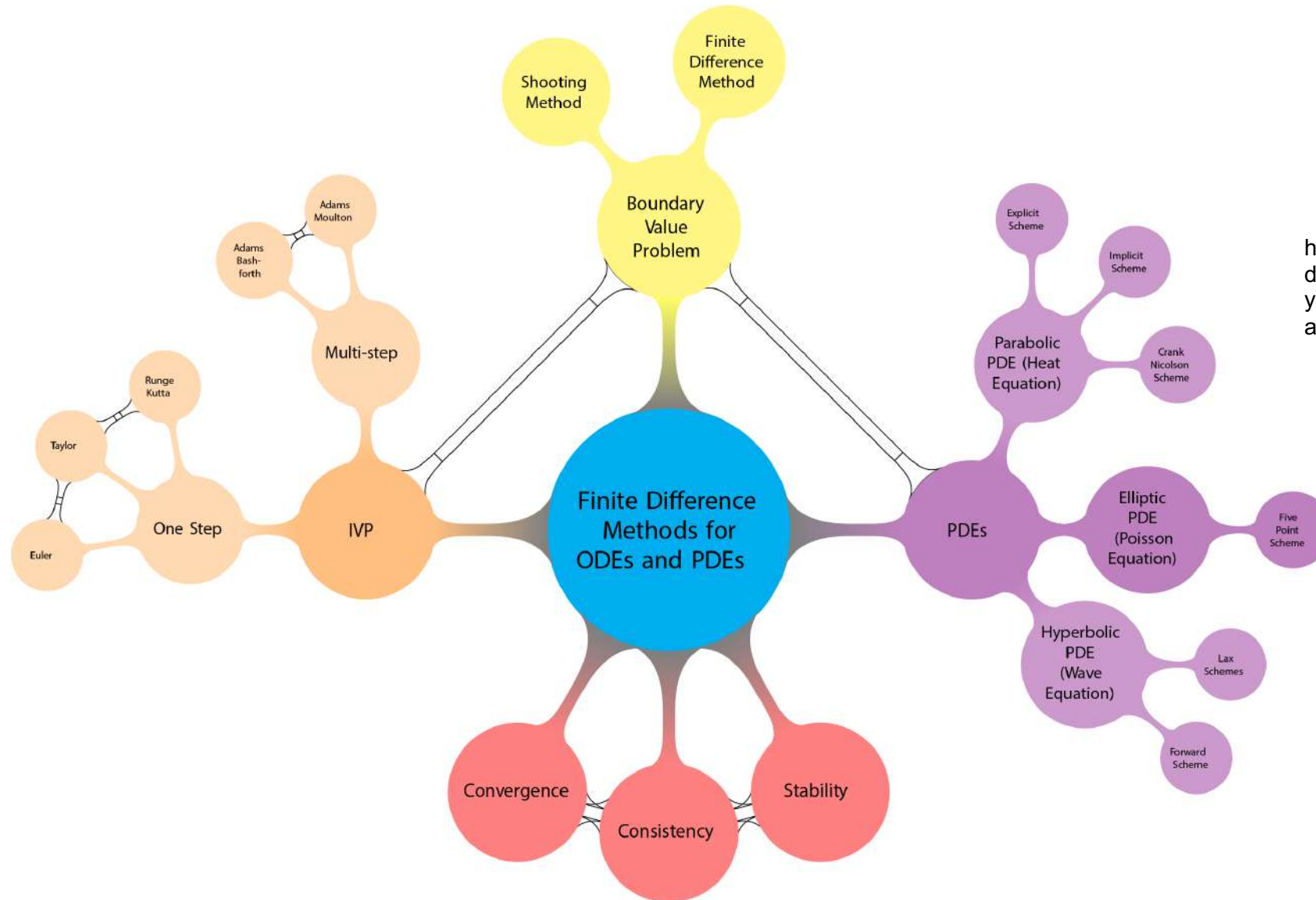
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Today's Lecture

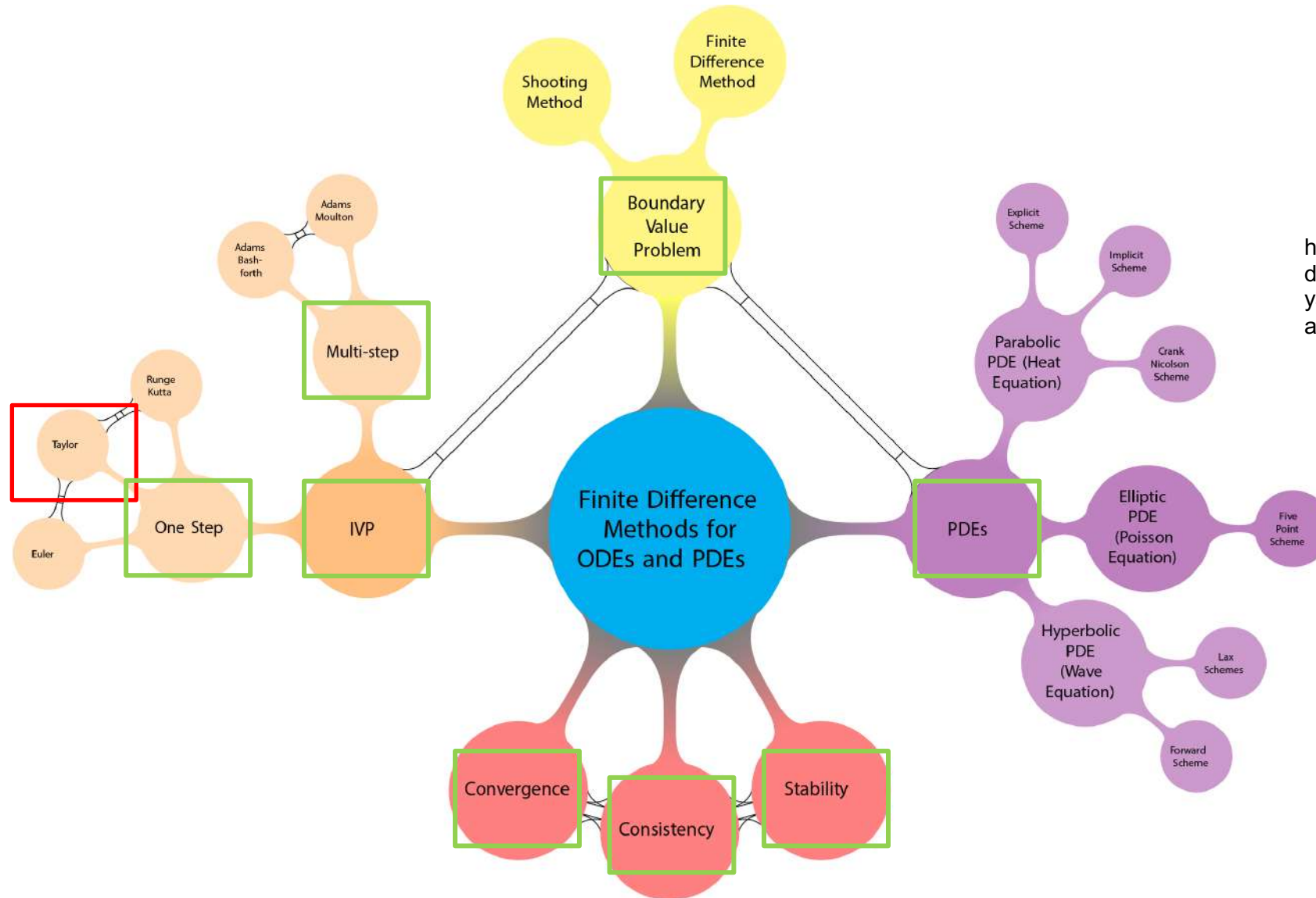
- Numerical Modelling Methods
- Gitlab and Github examples and getting it running
- Shallow Water Equations + Project assignment!

Introduction to Numerical Methods



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Introduction to Numerical Methods



https://john-s-butler-dit.github.io/NumericalAnalysisBook/_images/NumericalAnalysis.png

Ordinary Differential Equations

- An Ordinary Differential Equation (ODE) is an equation that involves an unknown function of a single variable, an independent variable as well as one or more of its derivatives.
- They were first considered by Leibnitz and Newton in the seventeenth century and are important throughout science, economic and engineering disciplines.
- In Numerical Modelling Applications, ODE are important for:
 - Orbit determination
 - Gravity field modelling
 - Ocean tide and load modelling
 - Earth deformation
 - Rotation

Ordinary Differential Equations

- An Ordinary Differential Equation (ODE) is an equation that involves an unknown function of a **dependent variable**, an **independent variable** as well as **one or more of its derivatives**.
- The general form of an ODE is given as:

$$F(x, y, y', y'', \dots y^{(n)}) = 0$$

- Where, **x is the independent variable**, **y is the dependent variable**, and the derivatives given by $y', y'', \dots y^{(n)}$.
- F is the function that relates the different variables.
- ODE's can be represented by different orders, with this equation representing the highest order (n).

Ordinary Differential Equations

- Examples of First-order ODE:

$$\frac{dy}{dx} = 2x$$

- Second-order ODE:

$$\frac{d^2y}{dx^2} + y = 0$$

- A simplified ODE form of a fluid flow:

$$\frac{dh}{dt} = \frac{Q_{in}(t) - Q_{out}(t)}{A}, \quad \text{where } Q_{out}(t) = C\sqrt{h(t)}$$

Is this an ODE?

Do we have the dependent variable, the independent variable and the derivative?

Ordinary Differential Equations: Example

$$\frac{dh}{dt} = \frac{Q_{in}(t) - Q_{out}(t)}{A}, \quad \text{where } Q_{out}(t) = C\sqrt{h(t)}$$

- This formula describes the basic in and out flow of fluid from a system. A much simplified version of the semester project we will do later.
- Let's solve for h assuming a fixed Q_{in} of 2, A of 100 m², C is constant at 1 and $h(0) = 0$. Solve for $t = 10$.
- How can we solve this equation?

ODE: Initial Value Problems

- In our previous example, the issue we had is we have an unknown in that we do not know what $h(t)$ is equal to. But we are given the initial state $h(0) = 0$.
- This is an initial value problem, which is when a solution to your ODE is given at some point within a domain or at a time.
- Typically these IVPs are solved using **analytical approaches** aimed at providing exact answers to ODE and using **numerical approaches** which use several approximations to solve these ODE.
- Both approaches have strengths and weaknesses, analytical approaches are usually precise but are not always possible (due to nonlinear characteristics etc).
- Numerical approaches give us algorithms or models that are based on the available information, but typically rely on making assumptions based on the solution being either impossible or too challenging (computationally) to get.

Initial Value Problems: Euler Method

- Eulers method is generally slightly outdated and rather simplistic.
- It is basically a tangential line approximation method with course correction over time.
- Take the example $y = x^2$, which has the differential equation of $\frac{dy}{dx} = 2x$. How do we solve for y at $x=1.5$, considering step sizes (h) of 0.1 considering that $x_o = 1$?
- Eulers equation is: $y_{n+1} = y_n + hf'(x_n, y_n)$.
- Our example of a differential equation in terms of x, we can rewrite this to $y_{n+1} = y_n + hf'(x_n)$. To then calculate y (1) we can go:

$$y_1 = y_o + hf'(x_o)$$

$$y_1 = 1 + 0.1 \frac{dy}{dx}$$

$$y_1 = 1 + 0.1 \times 2x_o$$

$$y_1 = 1.2$$

Initial Value Problems: Euler Method

- We can then expand by stepping x forward with our known h on this to get the following table:

n	x	y
0	1	1
1	1.1	1.2
2	1.2	1.42
3	1.3	1.66
4	1.4	1.92
5	1.5	2.2

- Therefore, our answer using Eulers method for solving y (1.5) would be 2.2.
- But how well does this compare to the analytical solution? $y = x^2$?

Initial Value Problems: Euler Method

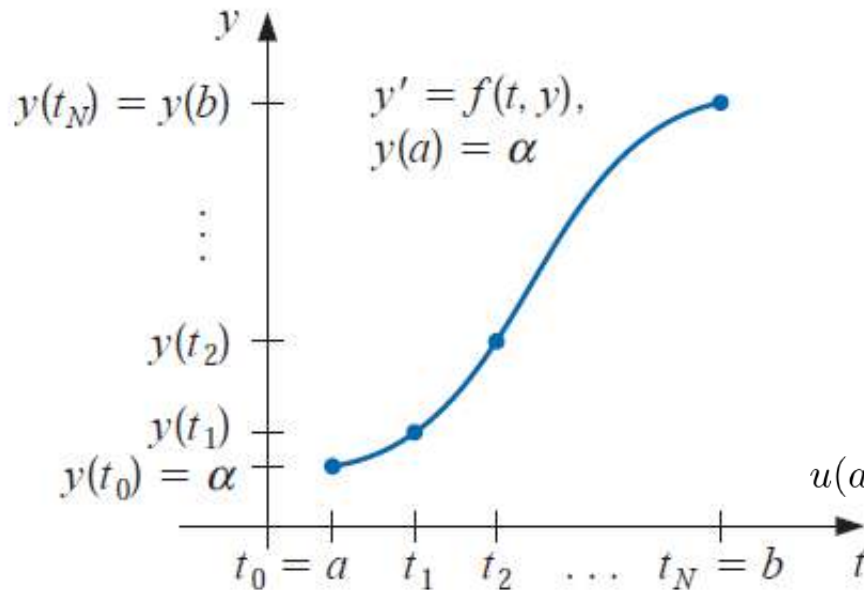
- We can then expand on this to get the following table ($y = x^2$):

n	x	y	Actual
0	1	1	1
1	1.1	1.2	1.21
2	1.2	1.42	1.44
3	1.3	1.66	1.69
4	1.4	1.92	1.96
5	1.5	2.2	2.25

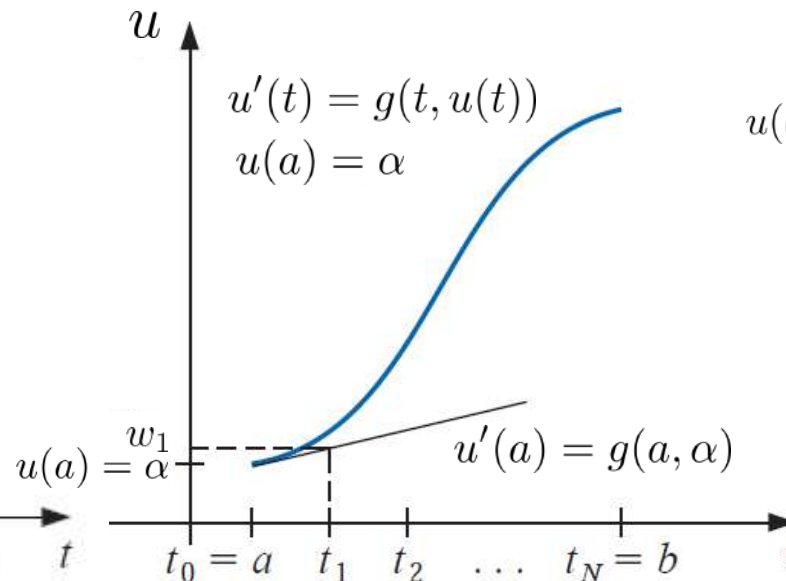
- We can see that the approximation of Eulers Method is okay, and not that far off the actually analytical solution of this equation.
- It is also quite obvious that if you increase the step size (which can be spatial or temporal resolution), your accuracy will increase. This of course then becomes a trade off with computational efficiency.
- Compared to a tangential line approach where you make one line estimate, Eulers method adjusts the estimation at each time step and therefore, makes a better estimation.

Initial Value Problems: Euler Method Visualized

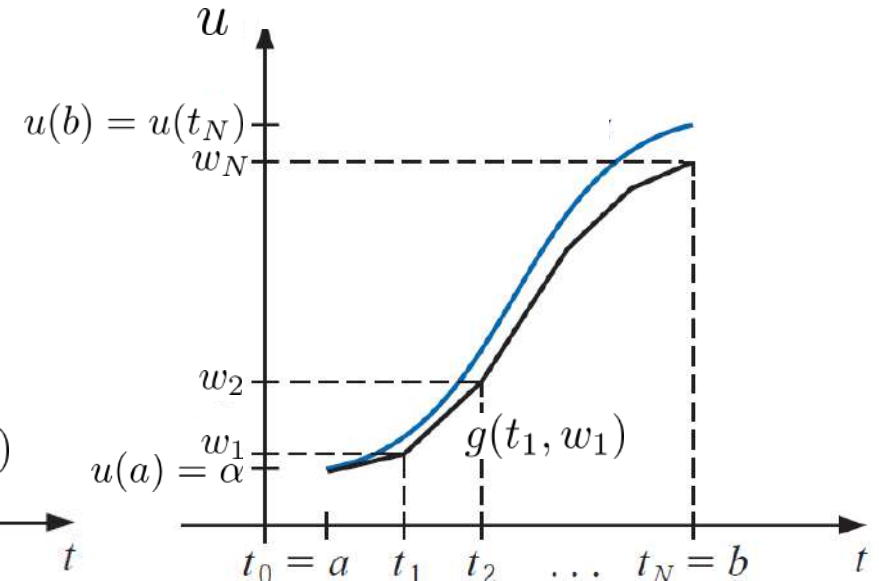
Solution



Tangential Estimation



Eulers Method



- The Euler method is the most **elementary approximation** technique for solving initial-value problems.
- Since the Euler method considers **only the first derivative**, i.e. the linear term in the Taylor expansion, it is also the **worst method** for solving an initial value problem

What about Truncation Error?

- The Euler method cuts of the Taylor series expansion at the first derivative, meaning that all higher order terms are neglected. The error caused by these neglected terms is called the Truncation Error.
- This happens particularly when an infinite process is cut off due to computational limitations and simplified calculations. An obvious example is **3.14** with pi, missing out several digits: **3.14159265359.....**
- In our Euler Example, we can then rewrite our equation of:

$$\begin{aligned} y_{n+1} &= y_n + hf'(x_n), \quad \text{to} \\ y_{n+1} &= y_n + hf'(x_n), \quad O(h^2) \end{aligned}$$

- With $O(h^2)$ being the local truncation error between each step, which can be obtained from the Taylor expansion to be: $\frac{h^2}{2} \cdot y''(\xi)$ which is explained in the Taylor expansion formula.
- If you want to see how to derive the Taylor expansion:
<https://courses.grainger.illinois.edu/cs357/fa2019/assets/lectures/Lecture8-Sept19.pdf>

Runge-Kutta Method

- The **Runge-Kutta methods** are an important family of **iterative methods** for the approximation of solutions of Ordinary Differential Equations (ODE).
- These methods have been **developed around 1900** by the German mathematicians
 - **Carl Runge** (1856 –1927) and
 - **Wilhelm Kutta** (1867 – 1944).
- The Runge-Kutta method is one of the most widely used ways to solve the initial value problem.
- This method can be used to derive high order accurate numerical models without needing higher order derivatives of functions.
- Unlike the Euler method, the Runge-Kutta methods uses a **weighted average of 4 different slopes**. This gives a **much better estimate** of where the curve is going.

Runge-Kutta Method

- Using the same approach in the Eulers example, $\frac{dy}{dx} = 2x$, the Runge-Kutta Fourth Order (RK4) method produces a model of $y_1 = y_0 + \frac{h}{6} (k_1 + 2k_2 + 2k_3 + k_4)$ which can be explained as follows:

$$f(x, y) = 2x$$

$$k_1 = f(x_n, y_n)$$

$$k_2 = f(x_n + \frac{h}{2}, y_n + \frac{h}{2} k_1)$$

$$k_3 = f(x_n + \frac{h}{2}, y_n + \frac{h}{2} k_2)$$

$$k_4 = f(x_n + h, y_n + hk_3)$$

- When we combine this we get: $y_1 = y_0 + \frac{h}{6} (k_1 + 2k_2 + 2k_3 + k_4)$.

Runge-Kutta Method Visualized

$$f(x, y) = 2x$$

$$k_1 = f(x_n, y_n)$$

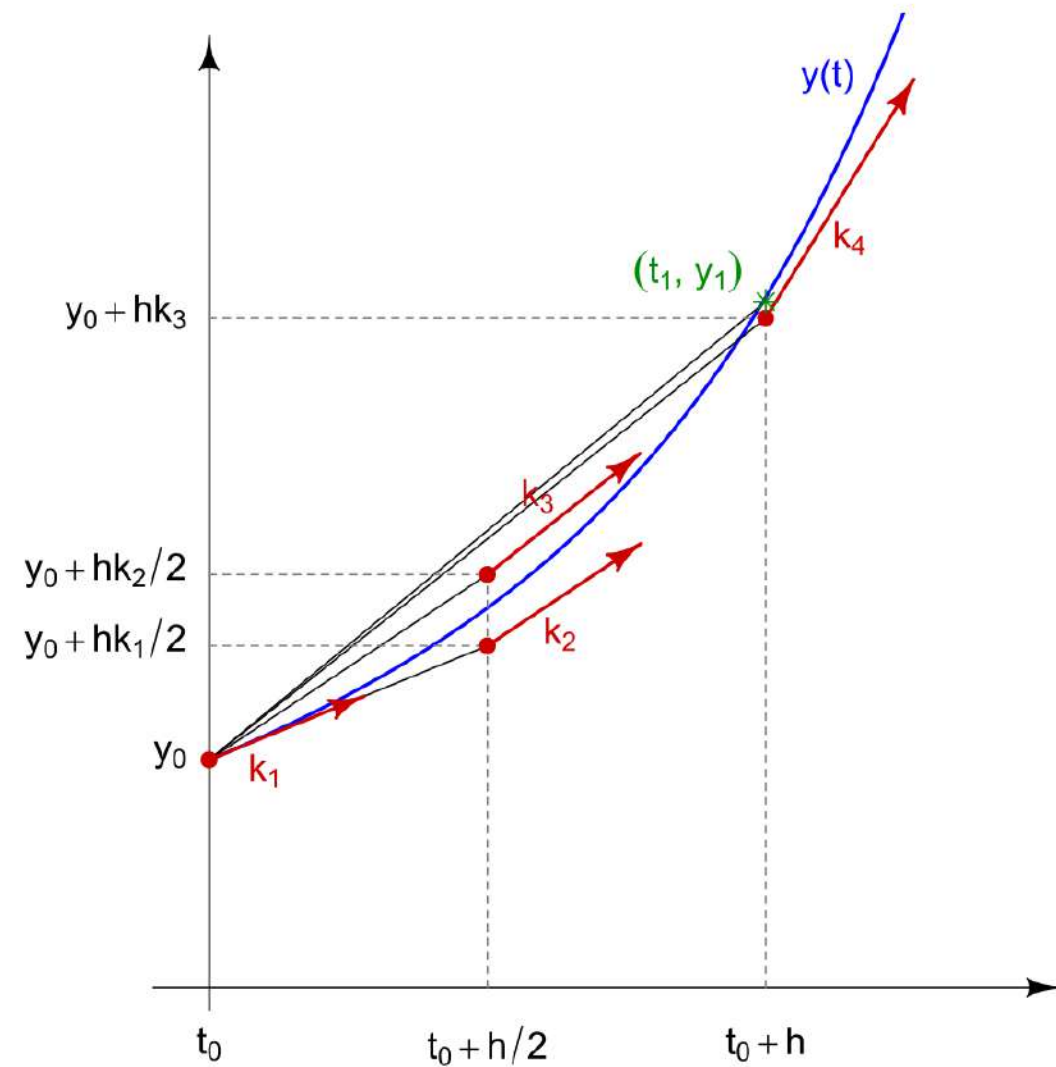
$$k_2 = f\left(x_n + \frac{h}{2}, y_n + \frac{h}{2} k_1\right)$$

$$k_3 = f\left(x_n + \frac{h}{2}, y_n + \frac{h}{2} k_2\right)$$

$$k_4 = f(x_n + h, y_n + hk_3)$$

$$y_1 = y_0 + \frac{h}{6} (k_1 + 2k_2 + 2k_3 + k_4).$$

Hiking a mountain analogy!



Initial Value Problems: Runge-Kutta

- We can then expand on this to get the following table ($y = x^2$):

n	x	y Euler	y RK4	Actual
0	1	1	1	1
1	1.1	1.2	1.21	1.21
2	1.2	1.42	1.44	1.44
3	1.3	1.66	1.69	1.69
4	1.4	1.92	1.96	1.96
5	1.5	2.2	2.25	2.25

- We can see that this approximation matches the actual analytical solutions perfectly, and clearly outperforms the Euler approach.
- The Runge-Kutta approach is a good trade-off between computation efficiency, complexity and model performances.

Single-step methods vs. Multi-step methods

- All the methods which have been presented before are for solving the **initial value problem** are so-called **single-step methods**. Single-step methods only use the current values (x_n, y_n) .
- Multi-step methods also exist that use several previous points to solve for the next value (x_{n+1}, y_{n+1}) .
- Both methods have their own strengths and weaknesses.
- Single-step: often easier to implement and do not require the storing of multiple past values; but are can be less efficient as you do multiple slope calculations and less accurate [unless very small step sizes are used]/
- Multi-step: more efficient and achieve higher accuracy with less computation; but are more complex to implement and require more starting values which are often obtained from Runge-Kutta-4 methods.

Mid-point Method

- This method is a second-order Runge-Kutta method.
- The idea of this method is to compute the slope at the current point; take a half step forward to estimate what y would be at the midpoint; evaluate this slope at the midpoint and then use that slope to update the full step of the numerical solution.
- The formula can be expressed as follows:

$$y_{n+1} = y_n + h \cdot f\left(x_n + \frac{h}{2}, y_n + \frac{h}{2} \cdot f(x_n, y_n)\right)$$

Adams - Bashforth methods

- The **Adams-Bashforth methods** are alternatives to the **Runge-Kutta methods** for the approximation of solutions of Ordinary Differential Equations (ODE).
- These methods have been developed by **John Couch Adams** (1819–1892), who was a British mathematician and an astronomer.
- He developed these numerical techniques to approximate the solution of a fluid-flow problem posed by the British scientist **Francis Bashforth** (1819 – 1912) .
- The **Adams-Bashforth 4-step method** has an **accuracy** of the order $O(h^5)$.
- Besides the Adams-Bashforth 4-step method there also exist the
 - **Adams-Bashforth 2-step method** and the
 - **Adams-Bashforth 3-step method**.
- We won't go over the specifics of how each of these methods are derived, but the next slide shows the formulations for the example case.
- Generally, the application of an **Adams-Bashforth n-step method** requires that the first $n + 1$ derivatives of the target function $u(t)$ are **continuous**.

Adams - Bashforth methods

- Adams-Bashforth 2-step method:

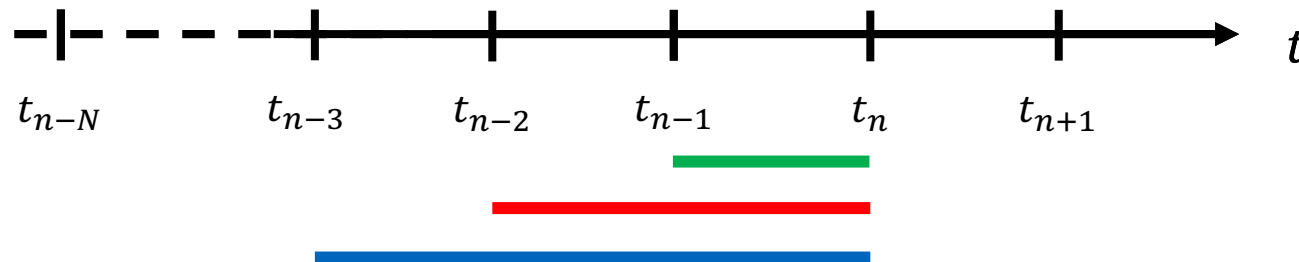
$$y_{n+1} = y_n + \frac{h}{2} (3f_n - f_{(n-1)})$$

- Adams-Bashforth 3-step method:

$$y_{n+1} = y_n + \frac{h}{12} (23f_n - 16f_{n-1} + 5f_{n-2})$$

- Adams-Bashforth 4-step method:

$$y_{n+1} = y_n + \frac{h}{24} (55f_n - 59f_{n-1} + 34f_{n-2} - 9f_{n-3})$$



Initial Value Problems: Adam-Bashforth

- We can then expand on this to get the following table ($y = x^2$):

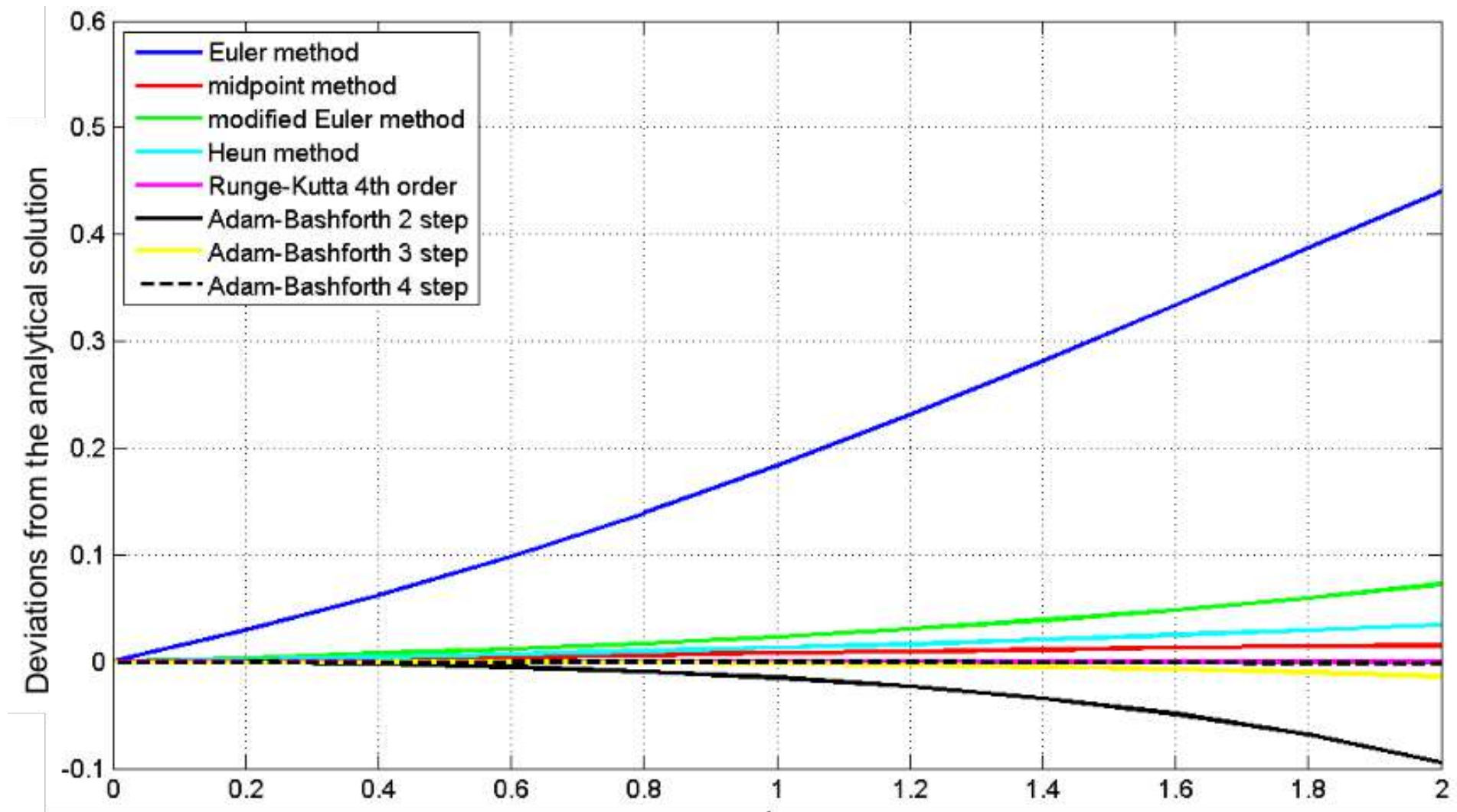
n	x	y Euler	y RK4	y AB4	Actual
0	1	1	1	1	1
1	1.1	1.2	1.21	1.21	1.21
2	1.2	1.42	1.44	1.44	1.44
3	1.3	1.66	1.69	1.69	1.69
4	1.4	1.92	1.96	1.96	1.96
5	1.5	2.2	2.25	2.25	2.25

- We can see that this approximation matches the actual analytical solutions perfectly, and clearly outperforms the Euler approach but aligns with the RK4 approach.
- The RK4 has benefits that it doesn't need the previous values, while AB4 of course relies a few previous values. The AB4 is more efficient as it does one call per step, rather than the 4 of the RK4.

A Summary of some Methods Used to Solve ODE

- **Midpoint method / Modified Euler Method:** This method is a second-order Runge-Kutta method. It derives a slope similar to the Euler method but instead of taking the beginning of the interval, this method attempts to get a more accurate estimate by evaluating derivative at the midpoint between time-steps.
- **Heun method:** This method uses the Euler approach to make a prediction of the next point, it then calculates a slope at the beginning as well as the end point. It then derives the mean of these two estimates to try get a more accurate estimation.
- **The Adam-Bashforth Method:** This is a linear multistep method used which uses past solution values to extrapolate and predict future solutions of an ODE.

A Summary Methods Used to Solve ODE



Ordinary Differential Equations: Example

$$\frac{dh}{dt} = \frac{Q_{in}(t) - Q_{out}(t)}{A}, \quad \text{where } Q_{out}(t) = C\sqrt{h(t)}$$

- This formula describes the basic in and out flow of fluid from a system. A much simplified version of the semester project we will do later.
- Lets code this:

$$Q_{in} = 2$$

Cross-section-Area = 100 m²

- How does the height look after 1800 seconds?
- What happens if we look at a CSA of 250m²?
- Doubling Q_{in} ?

Partial Differential Equations

- Partial Differential Equations (PDE) involves two or more independent variables and their partial derivatives, which is different to ODEs which is only one variable and its ordinary derivative.
- They are widely used in a variety of different applications ranging across mathematics, engineering and geophysics
- In Numerical Modelling Applications, PDE are important for:
 - Gravity field modelling
 - Shallow water equations (for sea level and ocean circulation)
 - **Navier Stokes equations**
 - Heat and momentum conservation equations
 - Equations of motion

Partial Differential Equations

- Partial Differential Equations (PDE) involves two or more independent variables and their partial derivatives, which is different to ODEs which is only one variable and its ordinary derivative.
- The general form of PDE can be described by the following:

$$F \left(x_1, x_2, \dots, x_m, u, \frac{\partial u}{\partial x_1}, \frac{\partial u}{\partial x_2}, \dots, \frac{\partial^2 u}{\partial x_1^2}, \frac{\partial^2 u}{\partial x_1 \partial x_2}, \dots \right) = 0$$

- $u = u(x_1, x_2, \dots, x_m)$ is the unknown function of several independent variables.
- $\frac{\partial u}{\partial x_1}$ is the first order partial derivatives.
- $\frac{\partial^2 u}{\partial x_1 \partial x_2}$ are the higher-order partial derivatives.

Second order Partial Differential Equations

- A second order PDE is generally defined as:

$$F\left(x, y, u(x, y), \frac{\partial u(x, y)}{\partial x}, \frac{\partial u(x, y)}{\partial y}, \frac{\partial^2 u(x, y)}{\partial x^2}, \frac{\partial^2 u(x, y)}{\partial x \partial y}, \frac{\partial^2 u(x, y)}{\partial y^2}\right) = 0$$

- where the pair (x, y) of the variables x and y is restricted to the **2-D domain (area)** Ω , i.e. $(x, y) \in \Omega$.
- With this PDE called **linear** if it can be written as

$$a_{2,0} \cdot \frac{\partial^2 u(x, y)}{\partial x^2} + a_{2,2} \cdot \frac{\partial^2 u(x, y)}{\partial x \partial y} + a_{0,2} \cdot \frac{\partial^2 u(x, y)}{\partial y^2} + a_{1,0} \cdot \frac{\partial u(x, y)}{\partial x} + a_{0,1} \cdot \frac{\partial u(x, y)}{\partial y} + a_{0,0} \cdot u(x, y) = v(x, y)$$

where $a_{2,0}, a_{2,2}, a_{0,2}, a_{1,0}, a_{0,1}, a_{0,0}$ are **constant parameters** or **linear or non-linear functions** depending on the variables x and y ; $v(x, y)$ is a given function which we will set to zero to assume a homogeneous state.

Second order Partial Differential Equations

- With this PDE called **linear** if it can be written as

$$a_{2,0} \cdot \frac{\partial^2 u(x,y)}{\partial x^2} + a_{2,2} \cdot \frac{\partial^2 u(x,y)}{\partial x \partial y} + a_{0,2} \cdot \frac{\partial^2 u(x,y)}{\partial y^2} + a_{1,0} \cdot \frac{\partial u(x,y)}{\partial x} + a_{0,1} \cdot \frac{\partial u(x,y)}{\partial y} + a_{0,0} \cdot u(x,y) = 0$$

- Now we assume the parameters $a_{2,0}, a_{2,2}, a_{0,2}, a_{1,0}, a_{0,1}, a_{0,0}$ are constants and obtain with $a_{2,0} = a, a_{2,2} = b, a_{0,2} = c, a_{1,0} = d, a_{0,1} = e, a_{0,0} = f$ the **linear PDE of 2nd order**
- We can make these assumptions as we are simplifying these more complex processes to allow for easier computation. This assumption holds for systems that don't vary much in space (i.e. in very small regional studies) and when modelling Earth or ocean properties that are deemed the same everywhere (sort of averaging for example for density or stiffness).

Second order Partial Differential Equations Classification

- When we take into account this approximation, our formula simplifies to:

$$a. \frac{\partial^2 u(x,y)}{\partial x^2} + b. \frac{\partial^2 u(x,y)}{\partial x \partial y} + c. \frac{\partial^2 u(x,y)}{\partial y^2} + d. \frac{\partial u(x,y)}{\partial x} + e. \frac{\partial u(x,y)}{\partial y} + f. u(x,y) = 0$$

- In practice, these second-order PDEs can be classified based on their second order terms. These types of classifications are as follows:
 - Elliptic: steady-state systems such as gravity field
 - Parabolic: diffusion-type of processes such as Earth's mantle modelling or Glacial Isostatic Adjustment
 - Hyperbolic: wave-like motions such as seismic activity or tsunami modelling
- We perform these classifications to allow us to determine the appropriate method to solve these solutions as the requirements to do so and **what physical processes can be studied by each equation**, more on this later.

Second order Partial Differential Equations Classification

$$a \cdot \frac{\partial^2 u(x,y)}{\partial x^2} + b \cdot \frac{\partial^2 u(x,y)}{\partial x \partial y} + c \cdot \frac{\partial^2 u(x,y)}{\partial y^2} + d \cdot \frac{\partial u(x,y)}{\partial x} + e \cdot \frac{\partial u(x,y)}{\partial y} + f u(x,y) = 0$$

- To do this classification on the above equation is done based on the discriminate of the second-order part:

$$D = b^2 - 4ac$$

- In dependence on the numerical value for the discriminant δ we distinguish between:
 - an ellipse ($D < 0$) or circle;
 - a parabola ($D = 0$);
 - a hyperbola ($D > 0$).
- **NOTE: some textbooks define the discriminate as $D = 4ac - b^2$ which would result in opposite signs in the above classification (i.e. ellipse would be $D > 0$)!**

Second order Partial Differential Equations Classification

$$a \cdot \frac{\partial^2 u(x,y)}{\partial x^2} + b \cdot \frac{\partial^2 u(x,y)}{\partial x \partial y} + c \cdot \frac{\partial^2 u(x,y)}{\partial y^2} + d \cdot \frac{\partial u(x,y)}{\partial x} + e \cdot \frac{\partial u(x,y)}{\partial y} + f u(x,y) = 0$$

- To simplify this explanation, we have assumed a homogenous state in the above equation $v(x,y)=0$ (i.e. the 0 on the right side of the equation) and rewrite the partial derivatives in their compact state. Our formula then becomes:

$$a u_{xx} + b u_{xy} + c u_{yy} + d u_x + e u_y + f u = 0$$

- Which can be explained by:

$$\begin{aligned} \bullet \quad u_{xx} &= \frac{\partial^2 u(x,y)}{\partial x^2} & u_{xy} &= \frac{\partial^2 u(x,y)}{\partial x \partial y} \\ \bullet \quad u_{yy} &= \frac{\partial^2 u(x,y)}{\partial y^2} & u_x &= \frac{\partial u(x,y)}{\partial x} \\ & & \bullet \quad u_y &= \frac{\partial u(x,y)}{\partial y} \end{aligned}$$

Second order Partial Differential Equations Classification

$$au_{xx} + bu_{xy} + cu_{yy} + du_x + eu_y + fu = 0$$

- By linear transformation of the above equation can be reduced as follows:

1. Elliptic Case

$$u_{xx} + u_{yy} + \dots = 0$$

2. Parabolic Case

$$u_{xx} - u_{yy} + \dots = 0$$

3. Hyperbola Case

$$u_{xx} + \dots = 0$$

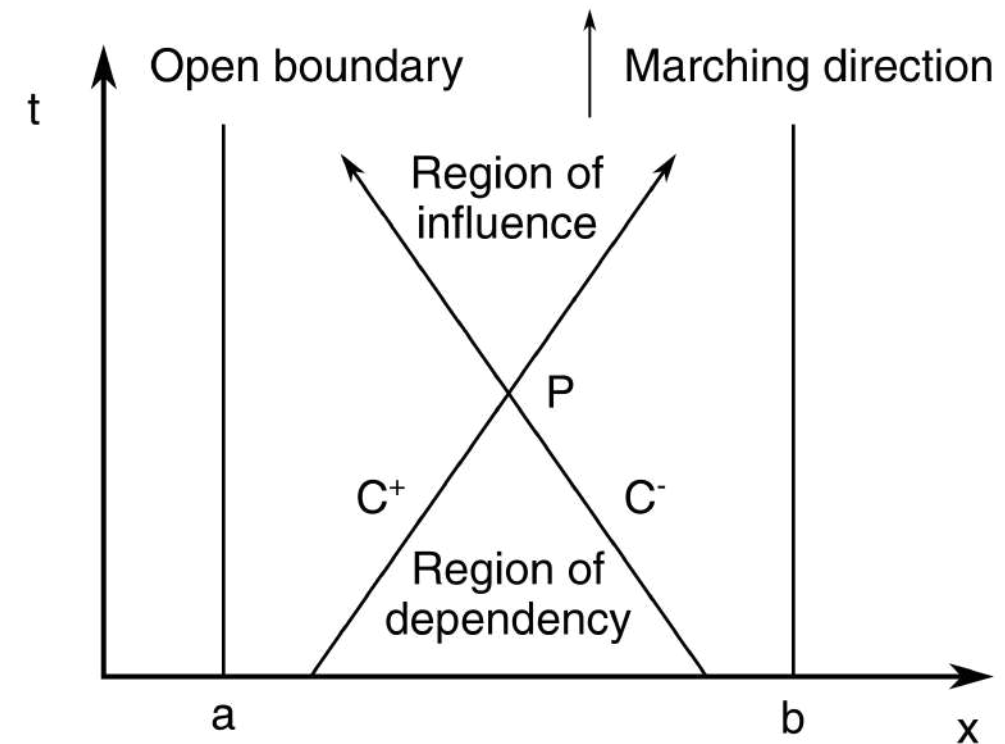
Where the ... represents the lower-order terms from the above equation.

Hyperbolic PDEs

- The most common example is the wave equation:

$$\frac{\delta^2 u}{\delta t^2} = c^2 \nabla^2 u$$

- Hyperbolic PDEs require information, as demonstrated by the wave equation, of the initial conditions for the time derivative as well as the function.
- Based on this, it is an initial value problem.
- This can be seen by the schematic, the position **P** is only dependent on the **Region of dependency**. But **P** only influences the **Region of influence**.

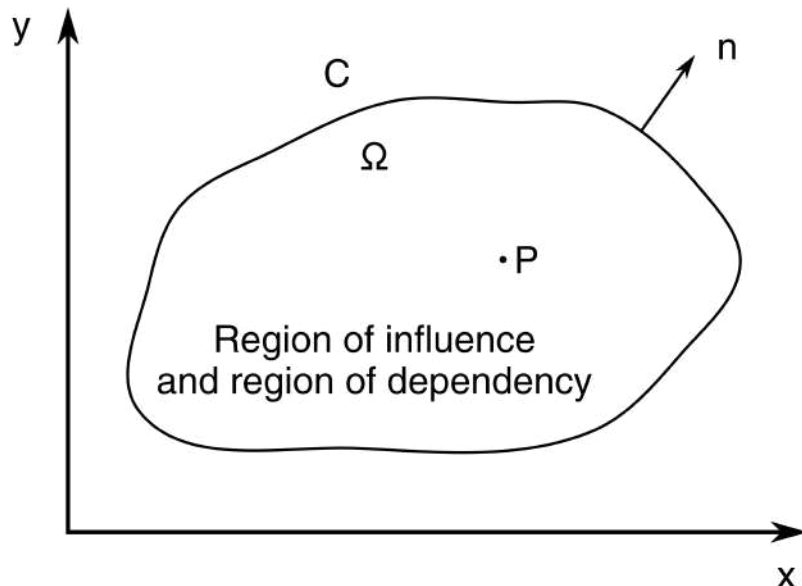


Elliptic PDEs

- Two common examples are Laplace's equation and Poisson's equation.

$$\text{Laplace Equation: } \Delta^2 u = 0$$
$$\text{Poisson's Equation: } \Delta^2 u = f(x, y)$$

- They have no time dependence, they are steady-state problems.
- Based on this, they require boundary conditions and are therefore, boundary value problems.



Where Ω is the domain, P is a random position you would like to model and C is the boundary of the model (or written as $\delta\Omega$).

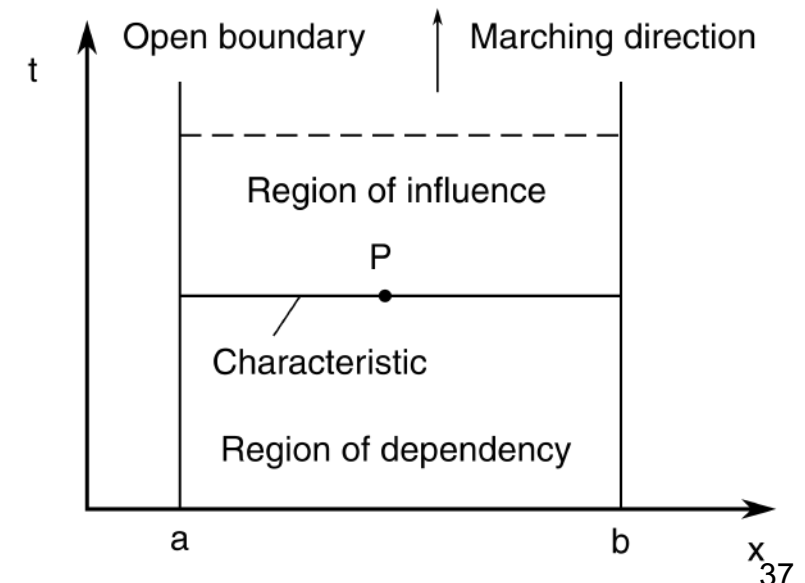
Parabolic PDEs

- The most common example is the Heat / diffusion equation:

$$\frac{\delta u}{\delta t} = \alpha \nabla^2 u$$

- They describe diffusion-like processes (processes that spread out smoothly with time) that involve both space and time and serve as a sort of inbetween hyperbolic and elliptic PDEs
- They require both an initial condition as well as boundary conditions and are described as initial-boundary problems.

The solution of P depends on all points for past time (t) points (**region of dependency**) and the present (influenced by boundaries).



Classification

Two of the most important examples of the above equations are the Poisson and Laplace Differential Equations.

- Poisson Differential Equation:

$$\frac{\partial^2 u(x,y)}{\partial x^2} + \frac{\partial^2 u(x,y)}{\partial y^2} = v(x,y)$$

- Which is reduced to the Laplace Differential Equation:

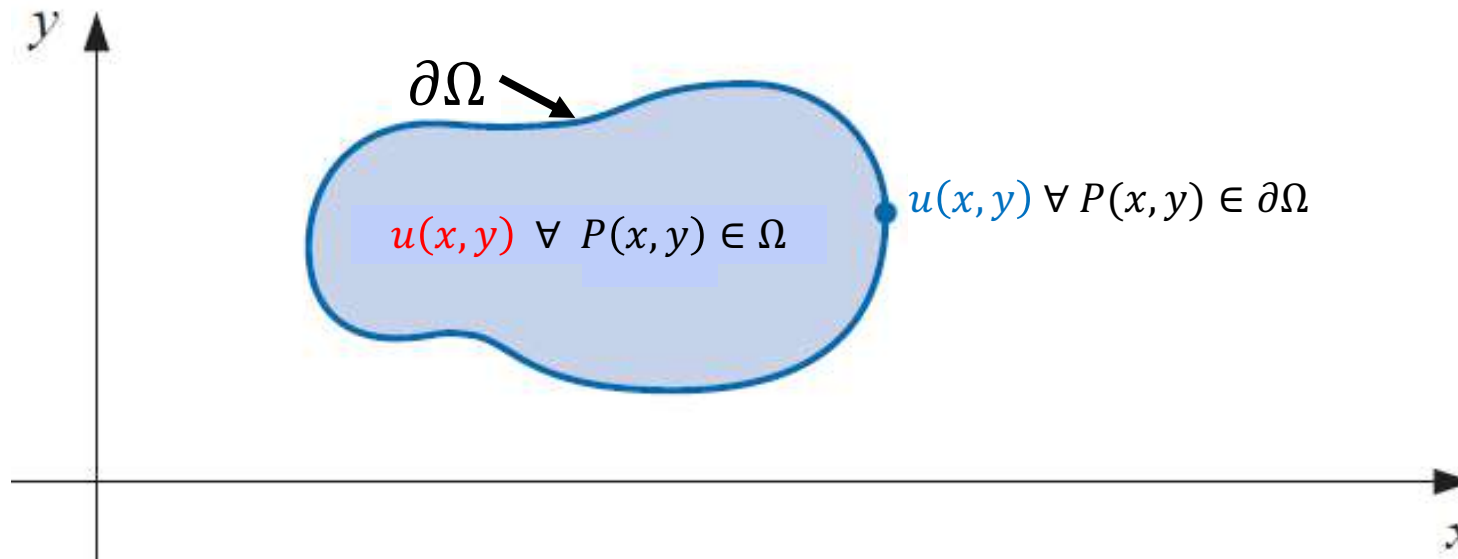
$$\frac{\partial^2 u(x,y)}{\partial x^2} + \frac{\partial^2 u(x,y)}{\partial y^2} = 0$$

- Both these equations are often used in **gravity field determination** (mostly in the 3-D case for the gravitational potential), in **electrostatics** and in other applications.
- Both of these equations are considered **elliptical PDEs**.

Poisson differential equation

$$\frac{\partial^2 u(x,y)}{\partial x^2} + \frac{\partial^2 u(x,y)}{\partial y^2} = v(x,y)$$

- To determine the unknown target function $u(x,y)$ within an area Ω fulfilling an elliptical PDE such as the Poisson equation, an inverse problem has to be solved.



- To get a solution for a 2D PDE (such as described above), i.e. a 2nd order differential equation, **boundary values** $u(x,y)$ along the **boundary curve** $\partial\Omega$ have to be given.

Boundary Conditions vs Boundary Value Problems

- These two concepts are related but are different by definition.
- Boundary value problems (which will be expanded on later), are PDEs where the solution is sought out within the a domain with specified boundary conditions.
- A boundary condition are the specific boundary conditions that are applied to the boundary value problem.
- So solving a boundary value problem requires boundary conditions which are introduced into the system using different approaches.

Boundary Conditions

$$\frac{\partial^2 u(x,y)}{\partial x^2} + \frac{\partial^2 u(x,y)}{\partial y^2} = v(x,y)$$

- We generally define for the solution of the 2nd order differential equation **three different kinds** of boundary conditions
- We distinguish between the
 - **Dirichlet condition**

$$u(x,y) = f(x,y) \quad \text{for} \quad (x,y) \in \partial\Omega$$

- $u(x,y)$ is the unknown solution of the PDE at point x,y .
- $f(x,y)$ is our known function
- $\partial\Omega$ represents the boundary of the domain Ω .
- If only Dirichlet boundary conditions are given, the problem to solve the 2nd order differential equation is called **Dirichlet boundary value problem**

Boundary Conditions

$$\frac{\partial^2 u(x,y)}{\partial x^2} + \frac{\partial^2 u(x,y)}{\partial y^2} = v(x,y)$$

- We generally define for the solution of the 2nd order differential equation **three different kinds** of boundary conditions
- We distinguish between the

- **Neumann condition**

$$\frac{\partial u}{\partial n}(x,y) = g(x,y) \quad \text{for} \quad (x,y) \in \partial\Omega$$

- $u(x,y)$ is the unknown solution of the PDE at point x,y .
- $g(x,y)$ is our known function specifying the the value of the normal derivative along the boundary.
- $\partial\Omega$ represents the boundary of the domain Ω .
- n is the outward unit normal vector to the boundary at the point x,y . $\frac{\partial u}{\partial n}$ is the normal derivative.

Boundary Conditions

$$\frac{\partial^2 u(x,y)}{\partial x^2} + \frac{\partial^2 u(x,y)}{\partial y^2} = v(x,y)$$

- We generally define for the solution of the 2nd order differential equation **three different kinds** of boundary conditions
- We distinguish between the
 - **Cauchy condition** (mixed conditions)

$$\frac{\partial u}{\partial n}(x,y) = g(x,y) \quad \text{for} \quad (x,y) \in \partial\Omega \quad \mathbf{AND} \quad u(x,y) = f(x,y) \quad \text{for} \quad (x,y) \in \partial\Omega$$

- This is a less common used approach in solving PDEs as it requires the definition of both conditions (normal derivative and solution at the boundary). It is more useful in hyperbolic PDEs compared to other approaches.

Boundary Conditions Visualized

- If we consider the 1D heat equation,

$$\frac{\partial u}{\partial t} = \alpha \cdot \frac{\partial^2 u}{\partial x^2}$$

- Dirichlet Condition Example:

$$T_0 = 0$$



$$T_L = 100$$

$$L = 10\text{m}$$

, where $u(0,t) = T_0$ and $u(L,t) = T_L$

- Neumann Condition Example (assuming no heat can flow in or out at $x=0$ and $x=L$):

$$\frac{\partial u}{\partial x}(0,t) = 0$$



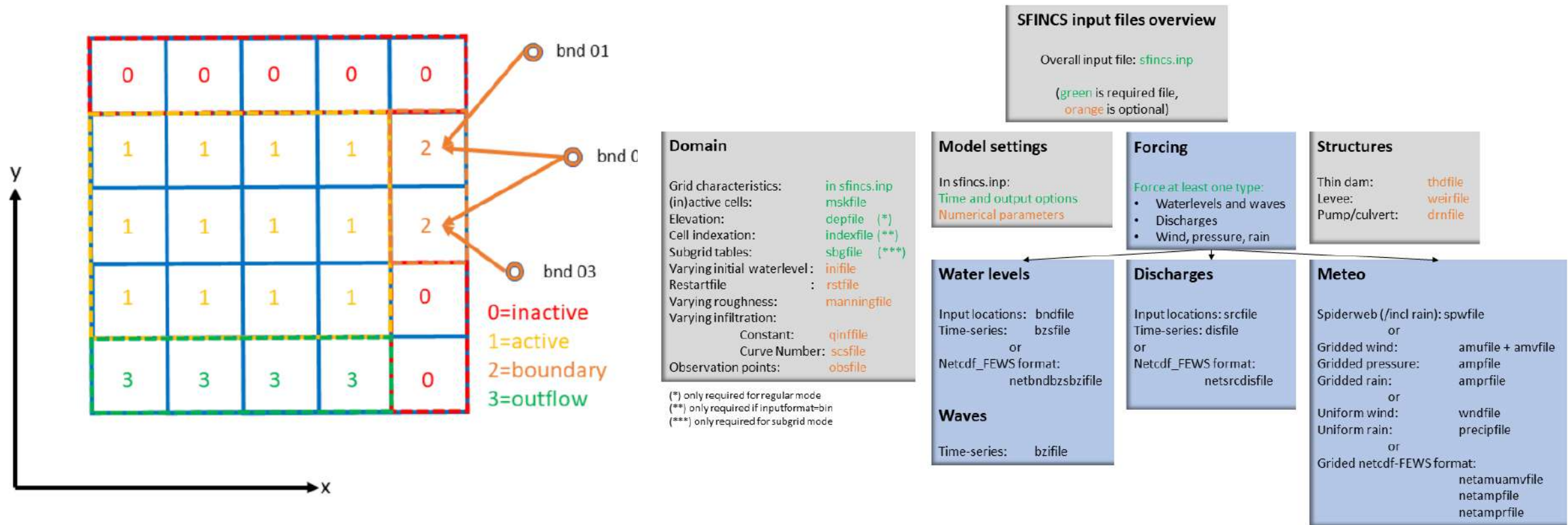
$$\frac{\partial u}{\partial x}(L,t) = 0$$

$$L = 10\text{m}$$

Re-life context of Boundary Conditions

- In many model studies, providing boundary conditions is a very common practice to account for certain phenomena that are not usually covered within the models.
- Through the incorporation of certain physics and then the propagation of their properties through the model we can improve our model physics.
- Applications include but are not limited to:
 - **Ocean models to incorporate:**
 - **Ocean tides**
 - Atmospheric processes
 - Incorporation of electromagnetic waves into ionospheric models
 - Heat transport into a system
 - Geoid determination

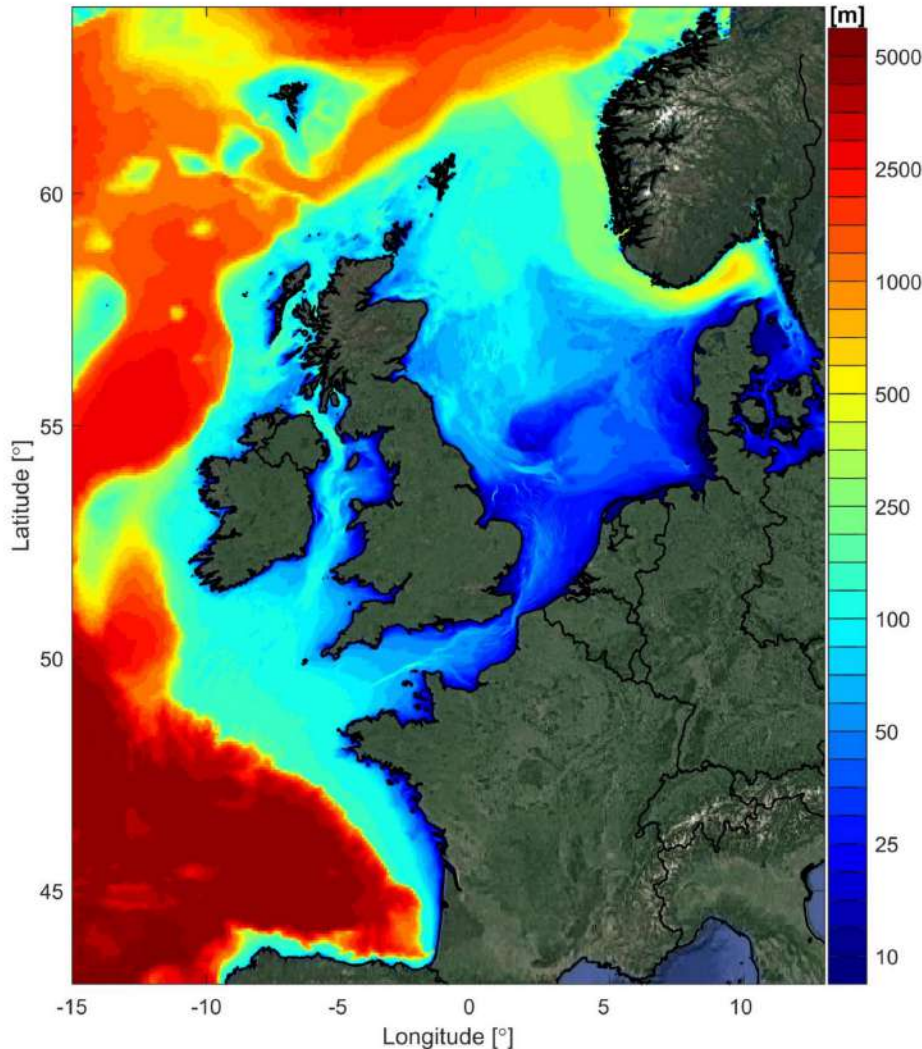
SFINCS: https://sfincs.readthedocs.io/en/latest/input_forcing.html



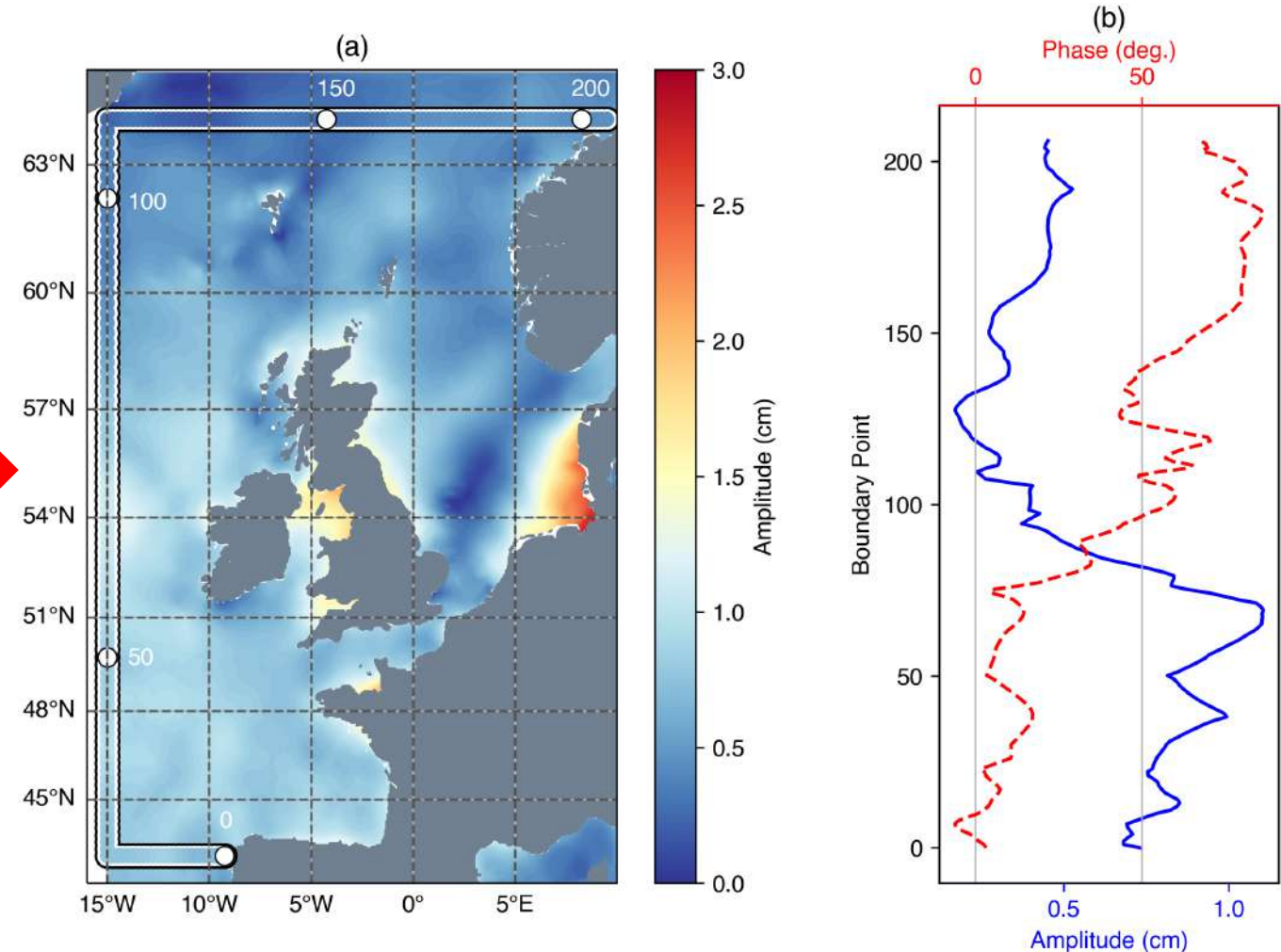
In this example for a flood prediction, the boundary forcings comes from 3 sources

Example of Boundary Forcing: Tides in Numerical Models

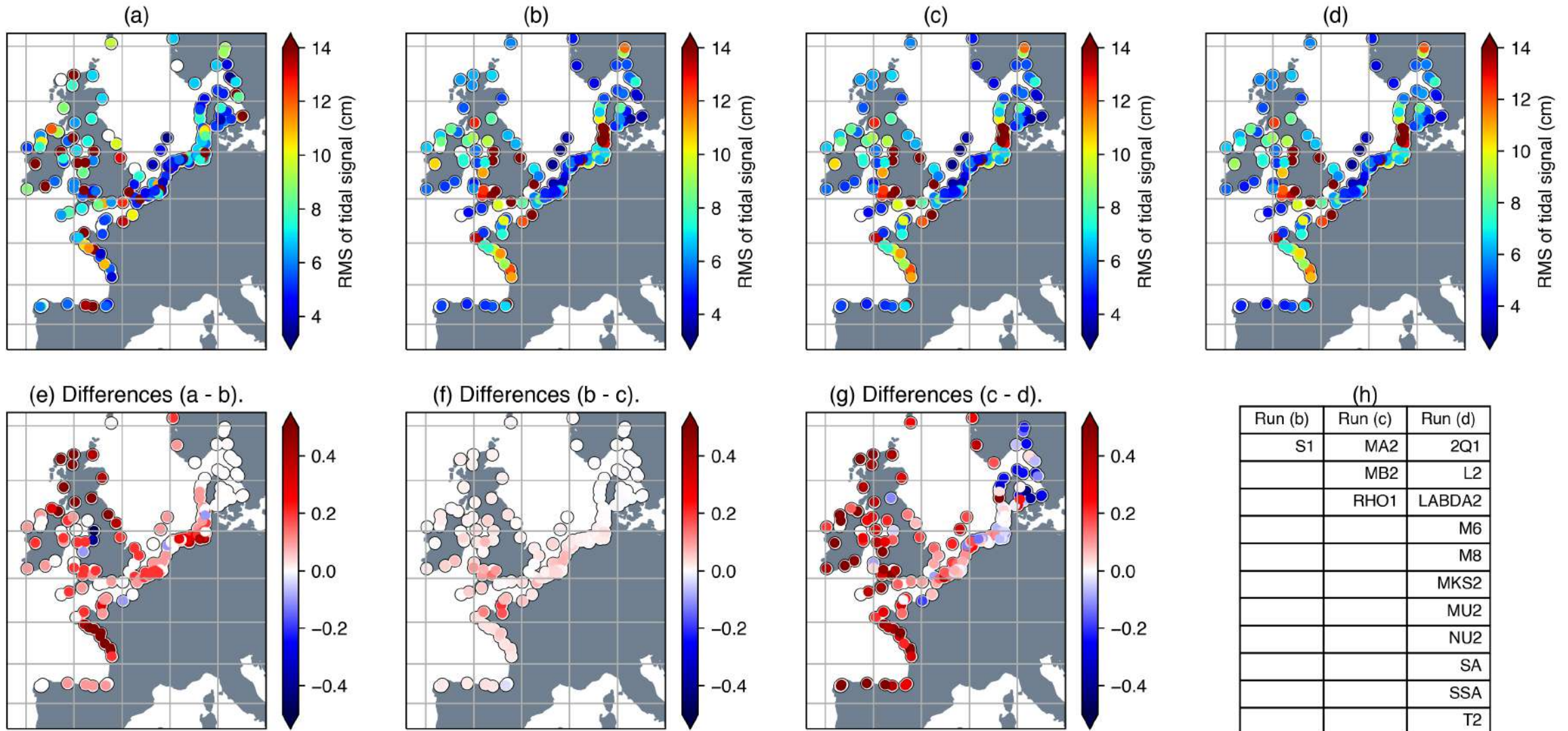
Dutch Continental Shelf Model



Empirical Ocean Tide Model



Example of Boundary Forcing: Tides in Numerical Models



Break

Boundary Value Problems

- Now that we have the boundary conditions, how do we know what is happening within the domain?
- To do this, we need to replace the derivatives with numerical approximations and use discrete points to turn our differential equation into a system of algebraic equations.
- Several methods exist to numerically solve ODEs and PDEs that have their own strengths and weaknesses. This includes (but are not necessarily limited to):
 1. Finite Difference Methods
 - a. Explicit Schemes
 - b. Implicit Schemes
 2. Finite Element Methods
 3. Finite Volume Methods
 4. Lagrangian Methods
 5. Spectral Methods
- BVPs require a spatial domain at least but can operate in both space and time.

BVP: Analytical Methods

- **[Rarely]** In some cases boundary value problems can be solved analytically, and this is of course the desired best result for solutions of processes.
- However, these are typically based on rather simple examples, like the analytical solutions of the ODEs in the last lecture.
- Here is a summary of a couple key methods:
- **Fourier Series and Eigenfunction Expansion:** Expresses the solution as an infinite series of eigen functions which is obtained by solving a **Sturm-Liouville BVP**. This approach also allows you to separate the natural modes of the system, allowing you study the components separately but also to get more complex solutions
- **Green's Functions:** Represents the solution as an integral using Green's functions that captures the influence of the point of source. Simply put, this approach tells you how a point responds to a pulse at a position [or boundary].

Finite Difference Methods

- The finite difference approximations for derivatives are one of the simplest and of the oldest methods to solve differential equations. It was already known by L. Euler (1707-1783) ca. 1768, in one dimension of space and was extended to dimension two by C. Runge (1856-1927) ca. 1908.
- The advent of finite difference techniques in numerical applications began in the early 1950s and their development was stimulated by the emergence of computers that offered a convenient framework for dealing with complex problems of science and technology.
- The principle of finite difference methods is close to the numerical schemes used to solve ordinary differential equations and solve boundary value problems.

Finite Difference Methods: Forward Difference

- A simple method to approximate a derivative using two nearby points is the forward difference approach. This is described by:

$$f'(x) = \frac{f(x + h) - f(x)}{h}$$

- In the **real world** we are dealing with discrete measurements which are characterized by a **finite sampling interval** (step size) h ,
- Depending on the **measurement system** it could be e.g. 1 sec, 30 sec, 1 hour, 1 day, 1 month, 1 year, but also 1 mm, 1 m, 5 km, etc.
- The measurement step size in time and space, is commonly referred to as temporal resolution and spatial resolution.
- It has an accuracy of the order $O(h)$ [i.e. the model accuracy improves with high resolution step sizes].

Finite Difference Methods: Backward Difference

$$f'(x) = \frac{f(x) - f(x - h)}{h}$$

- This approach is the inherently the same as the Forward Difference but taking the previous step, rather than the future step.
- It has the same accuracy order $O(h)$

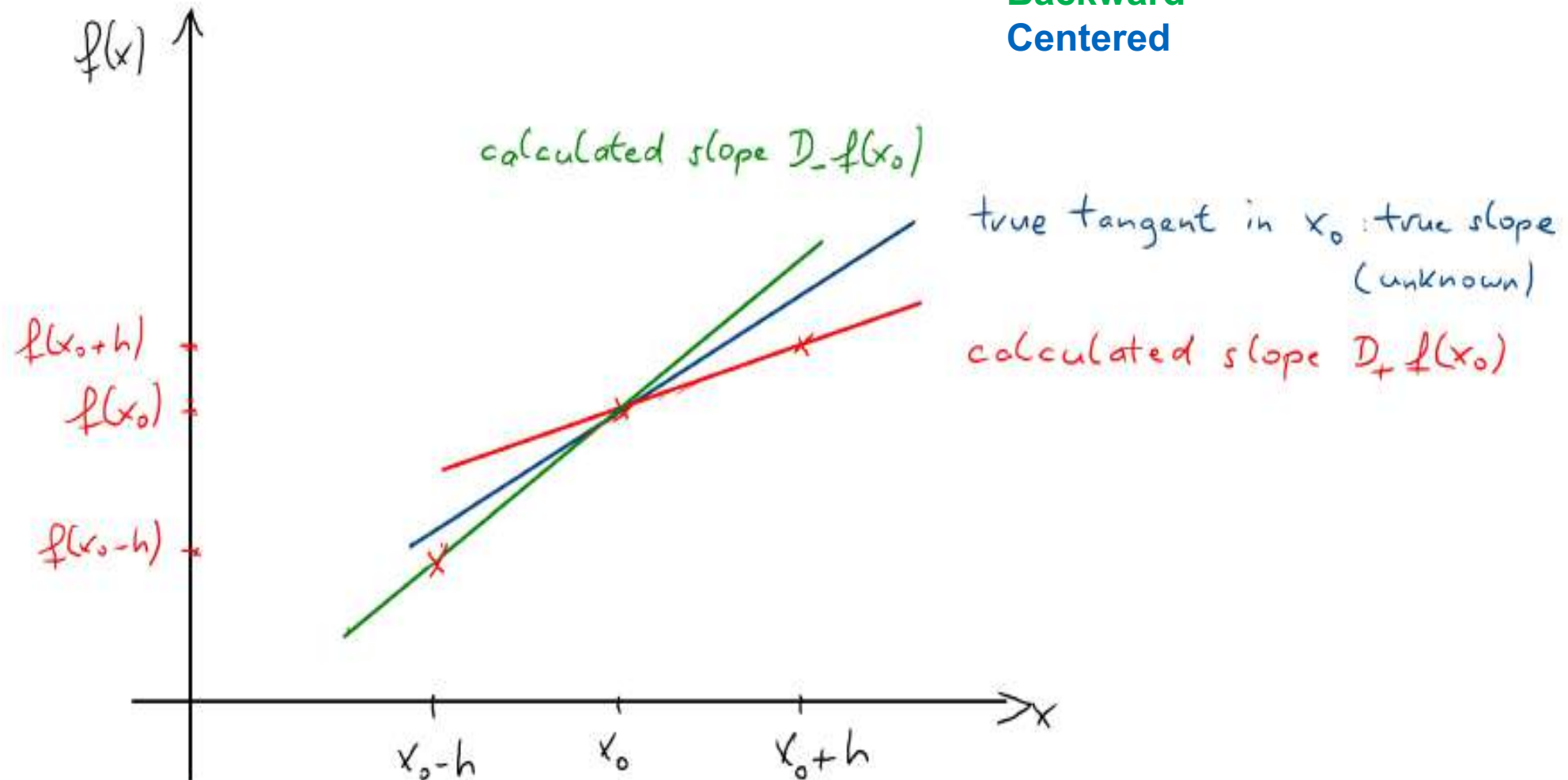
Finite Difference Methods: Centered Difference

$$D_0 f(x_0) = \frac{1}{2h} [f(x_0 + h) - f(x_0 - h)]$$

- This is a three-point approximation, as it considers the $x_0 - h$, x_0 and $x_0 + h$, and has a different accuracy order $O(h^2)$.
- The order of the accuracy can be increased by considering more points, logically, with four and five point approximations having accuracies of $O(h^3)$ and $O(h^4)$, respectively.

Finite Difference Methods

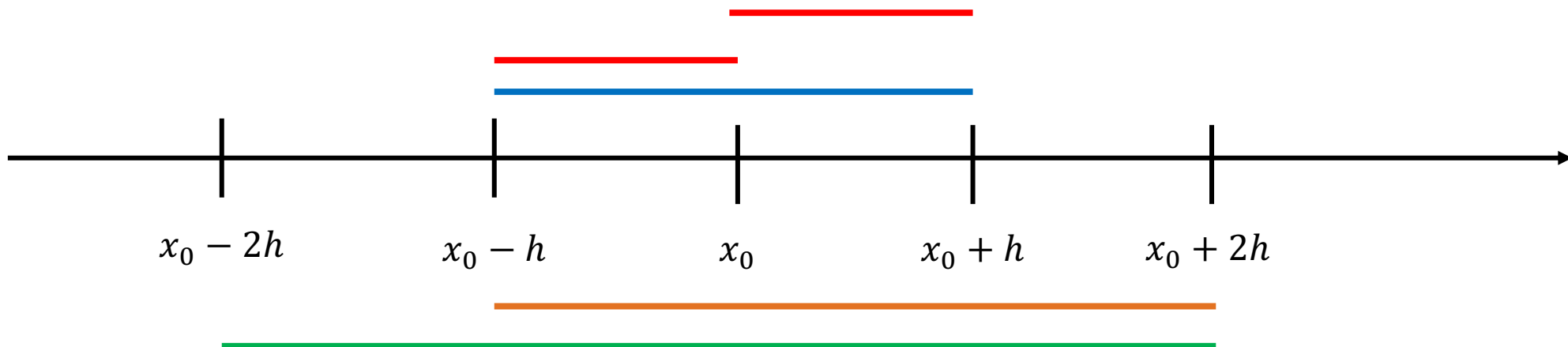
Forward
Backward
Centered



Finite Difference Methods

From the previous investigations we can summarize:

- **two-point formulae [e.g. Forward and Backward approaches]** are characterized by an **approximation accuracy of the order $O(h)$** and
- **three-point formulae [e.g. Centered Differences]** by an **approximation accuracy of the order $O(h^2)$**
- **four-point formulae** by an **approximation accuracy of the order $O(h^3)$**
- **five-point formulae** by an **approximation accuracy of the order $O(h^4)$**



Example: Laplace's Equation

- Lets use the Central Differences approach to discretize the Laplace's Equation. Recall, the discretization is the processes of converting differential equations into discrete forms that can be solved. I.e. they are the application of the learnt finite difference methods to real formulas.
- Laplace's Equation can be expanded to be: $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$, assuming equal grid sizing: $\Delta x = \Delta y$.
- The resultant discretization with the central differences methods would be:

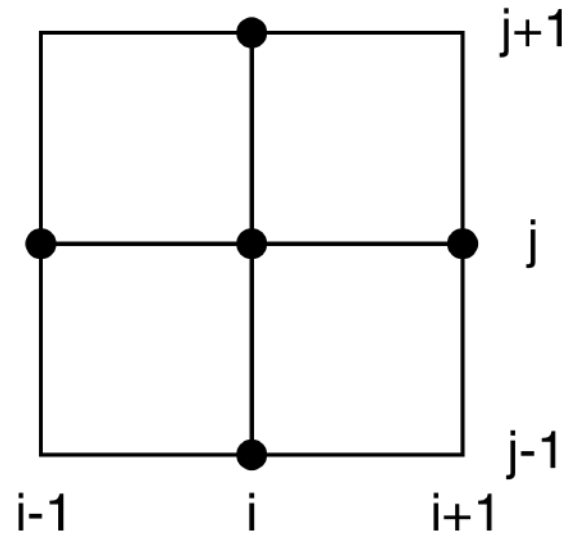
$$\frac{u_{i+1,j} - 2u_{i,j} + u_{i-1,j}}{\Delta x^2} + \frac{u_{i,j+1} - 2u_{i,j} + u_{i,j-1}}{\Delta y^2} = 0$$

- Multiply the equation by $h = \Delta x = \Delta y$, we are left with:

$$u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1} - 4u_{i,j} = 0$$

Example: Laplace's Equation

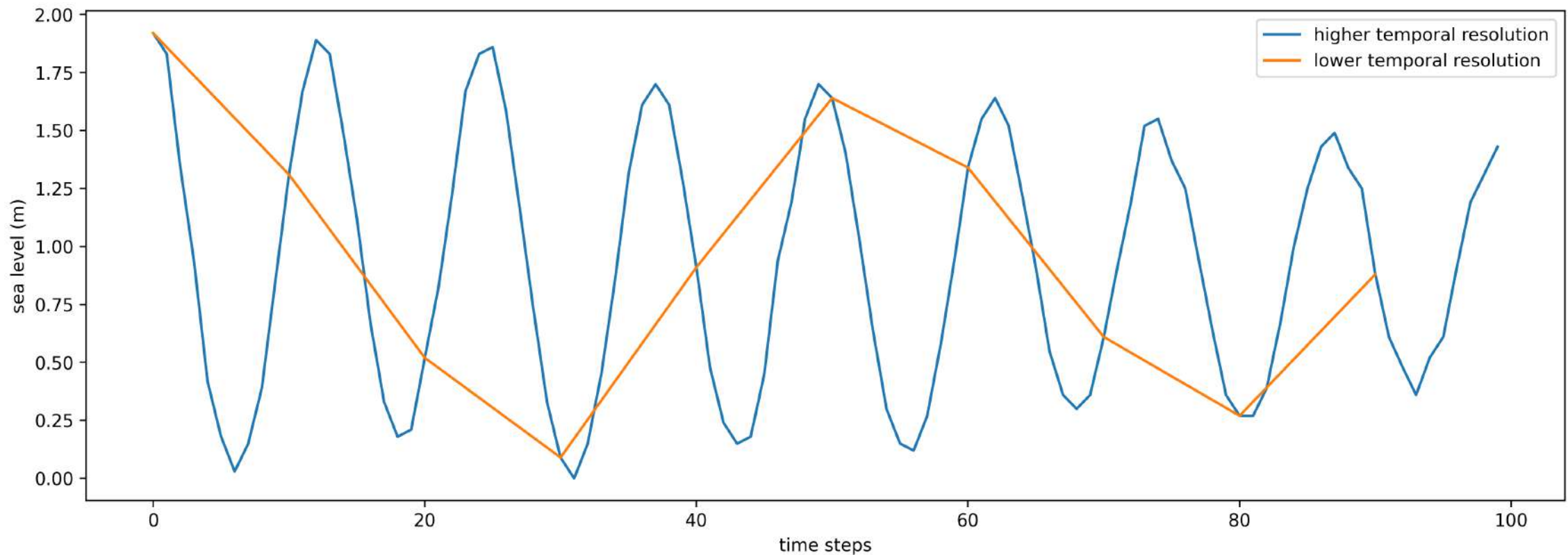
- This results in a well known 5-point numerical stencil (often termed the 5-point Formula) that can be illustrated as:



- When using the central differences approach in this case, an issue may arise when the location of this stencil is too close to the boundary of the computational domain (in this case spatial resolution may play an important role).
- In that case, you can replace the central differences with the backward or forward difference methods.

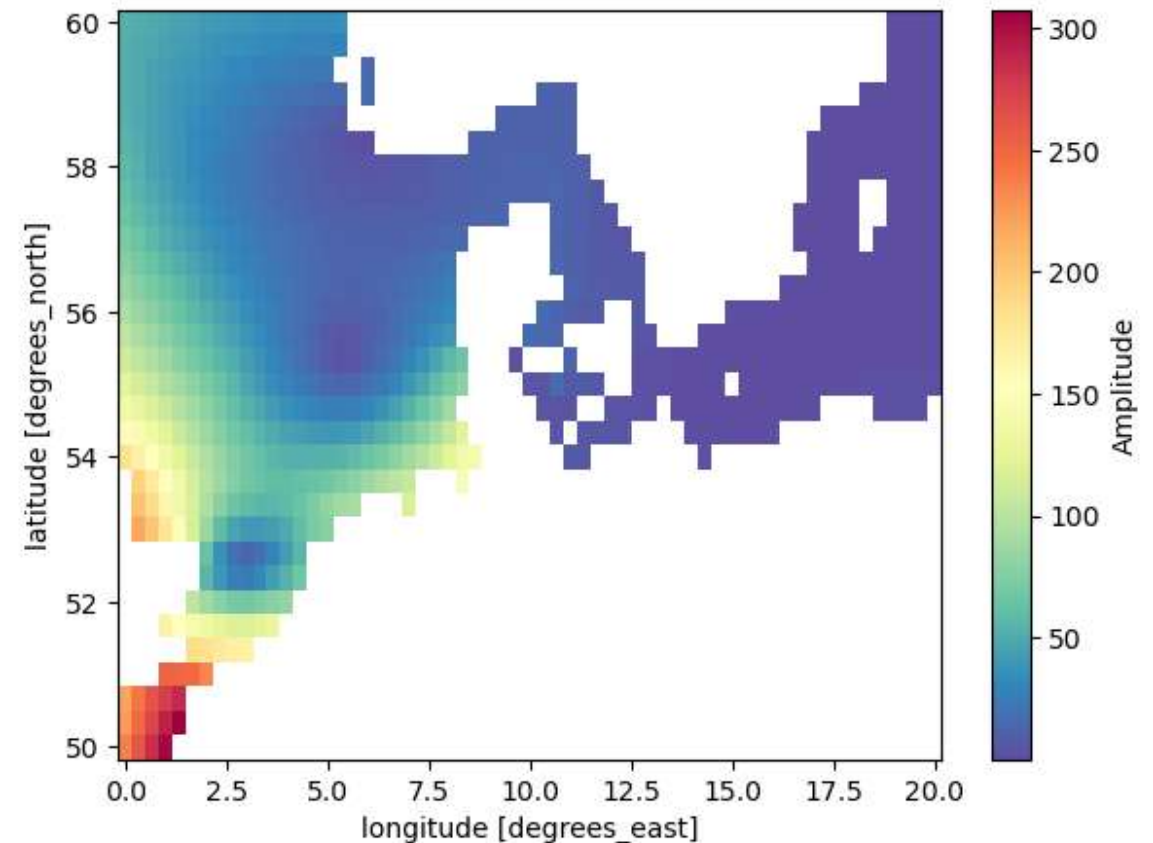
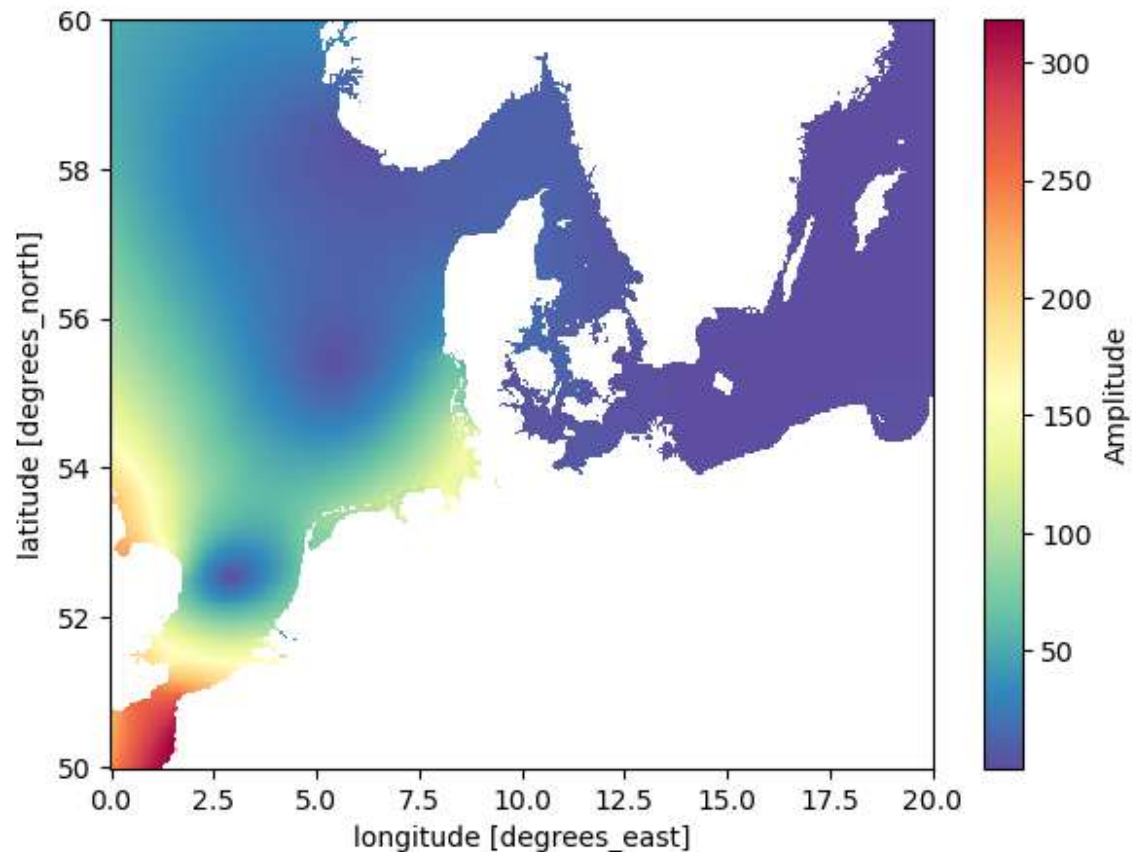
Side note on resolution

- The selection of this is important for model accuracy but also model stability. You cannot simply make this as small as possible [also computationally] as this will ruin the model stability.
- But for model accuracy, I show an example of why temporal resolution is important:

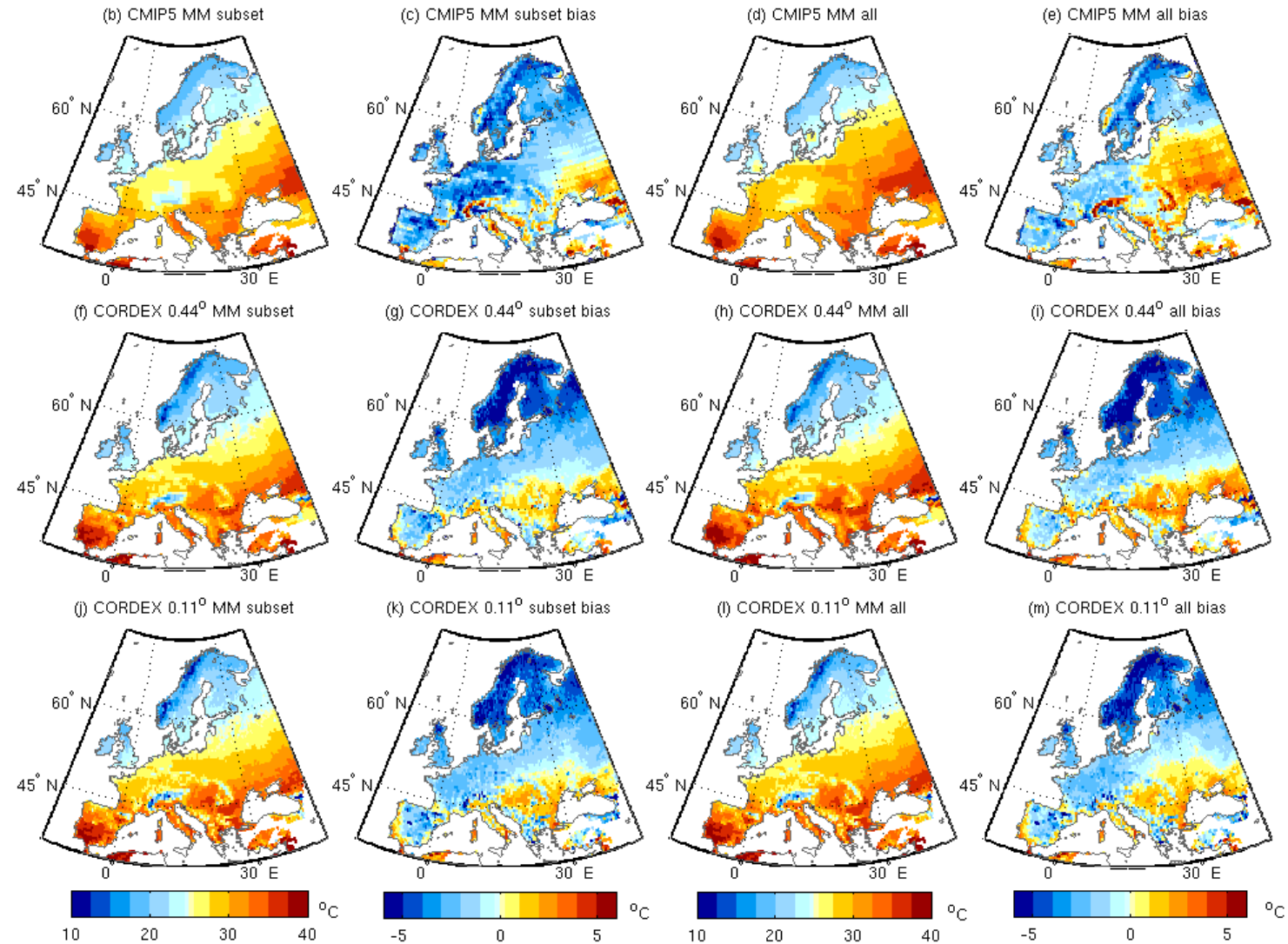


Side note on resolution

- Aside from stability, the cost of increasing the resolution is heavy on the computational time and load of your model.
- But for model accuracy, I show an example of why spatial resolution is important:



Effect of resolution affects all models, e.g. climate models:



<https://gmd.copernicus.org/articles/13/5583/2020/>

Important Considerations: Consistency, Stability, Convergence

- As we cannot resolve the differential equations analytically, at least not all the time, we have to discretize the equations (converting the differential equations into discrete forms that we can solve computationally).
- The error in a discretization is the difference between the solution of the original problem and the solution of the discrete problem, which must be defined so that the difference makes sense and can be quantified.
- **Consistency** of a discretization refers to a quantitative measure of the extent to which the exact solution satisfies the discrete problem.
- **Stability** means that the error of the solution does not increase unconditionally as the solution progresses with time.
- **Convergence** refers to the whether the solution approaches the exact solution as the model step (time and space) moves towards zero.

Stability: Short recap

- This was covered in introduction to Numerical modelling (for some of you, hopefully).
- The computation algorithm must be stable and not susceptible to any numerical problems arising from the accumulation and amplification of round-off errors.
- This means that any spontaneous perturbations of the solution may not have a tendency to be amplified at any point during the procedures. Otherwise, the computational errors get amplified and leak into the real solutions, making the results meaningless.
- This often puts a limitation on the temporal sampling, time steps.
- In numerical modelling and their applications, ensuring the stability of the selected numerical scheme is essential!

Stability: Short recap

- Early numerical models developed for atmospheric and oceanic applications, or weather predictions, especially the first presented by L.F. Richardson, did not consider the stability of the numerical solutions.
- Thus, although these models developed during the early 1920s were pioneering in every possible sense, they were not particularly useful. In the late 1920s, Courant, Friedrichs and Lewy were the first to consider numerical stability.
- The CFL (Courant-Friedrichs-Lewy) criterion: criterion dictates that the time step used in a numerical simulation must be small enough so that information (like a wave or signal) does not travel too far in a single time step. If this condition is not met, the simulation can produce unstable and inaccurate results.
- The CFL criterion essentially says that the distance the wave or particle travels in one time step (determined by its speed and the time step size) must be less than or equal to the grid spacing.

Mathematically, for a one-dimensional case with a wave speed $C = \frac{|u|\Delta t}{\Delta x}$.

Stability: Short recap

- The limit for stability within explicit numerical schemes is typically $C \leq 1$.
- This means the time step (Δt) must be chosen such that the product of the wave speed (c) and the time step (Δt) is less than or equal to the distance between grid points (Δx).
- The $C = \frac{|u|\Delta t}{\Delta x}$ describes a 1D solution while a 2D solution is represented by:

$$C = |u| \frac{\Delta t}{\Delta x} + |v| \frac{\Delta t}{\Delta y}.$$

- In essence, the CFL condition ensures that information in the numerical simulation does not propagate faster than it does in the physical system.
- The Von Neumann stability analysis is also a powerful tool to evaluate the stability of finite difference approximations [more on this later].

Numerical Solutions: Stability, Consistency and Convergences

- When we try to numerically solve challenging mathematical equations such as those governing fluid dynamics and motion, several important considerations are required:
 1. Does the numerical solution converge to the solution of the original differential equations when both time and space intervals approach zero? I.e. as resolution becomes finer, does the results become better.
 2. Is the numerical scheme stable, does it behave appropriately in time?
 3. How well do the numerical solutions match the exact solution in both amplitude (magnitude of the signal) and phases (timing of the signal).
- **We will keep these in mind when discussing different methods.**

Probably stop and move to Git?